



THERMOREGULATION

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Learning Outcomes:

- **Define** basal metabolic rate (BMR) and explain several factors that affect it.
- **Explain** Normal body temperature.
- **Describe** the factors that influence body heat production.
- **Explain** how normal body temperature is maintained by negative feedback loops involving the hypothalamic thermostat.

Body Temperature

- Normal Body Temperature (NBT) –
(36.1-37.2) C°
- Rectal Temp (a bit higher by 0.5 C°) reflects the internal body Temp (Core Body Temp)
- Core Body Temp (the temperature of the deep tissues of the body) remain almost constant
- Shell Temp (the temperature near the body surface in the skin and subcutaneous layer)-----Variable

Heat Balance

- Balance between heat production & heat loss (Heat Balance)
- Heat Balance
 - Heat production = Heat loss
- Heat production is called thermogenesis
- Heat loss is called as thermolysis

- The **Metabolic rate** is the overall rate at which metabolic reactions use energy, some of the energy is used to produce ATP, and some is released as heat.
- **Basal Metabolic Rate (BMR)**: is the measurement of the metabolic rate obtained under specific conditions (when the body is in a quiet, resting, and fasting condition).

Heat Production (Thermogenesis) continued...

The production of **body heat** is *proportional* to **metabolic rate**. Several factors affect the metabolic rate and thus the rate of heat production:

1. Exercise.
2. Hormones.
3. Nervous system.
4. Ingestion of food.
5. Age.
6. Other factors

Heat Production (Thermogenesis) continued...

- **Exercise.** During strenuous exercise, the metabolic rate may increase to as much as 15 times the basal rate.
- **Hormones.** Thyroid hormones (T3 and T4) are the main regulators of BMR; BMR increases as the blood levels of thyroid hormones rise. The response to changing levels of thyroid hormones is slow, however, taking several days to appear.
- **Nervous system.** During exercise or in a stressful situation, the sympathetic nervous system is activated that enhance the release of epinephrine and norepinephrine which they increase the metabolic rate.

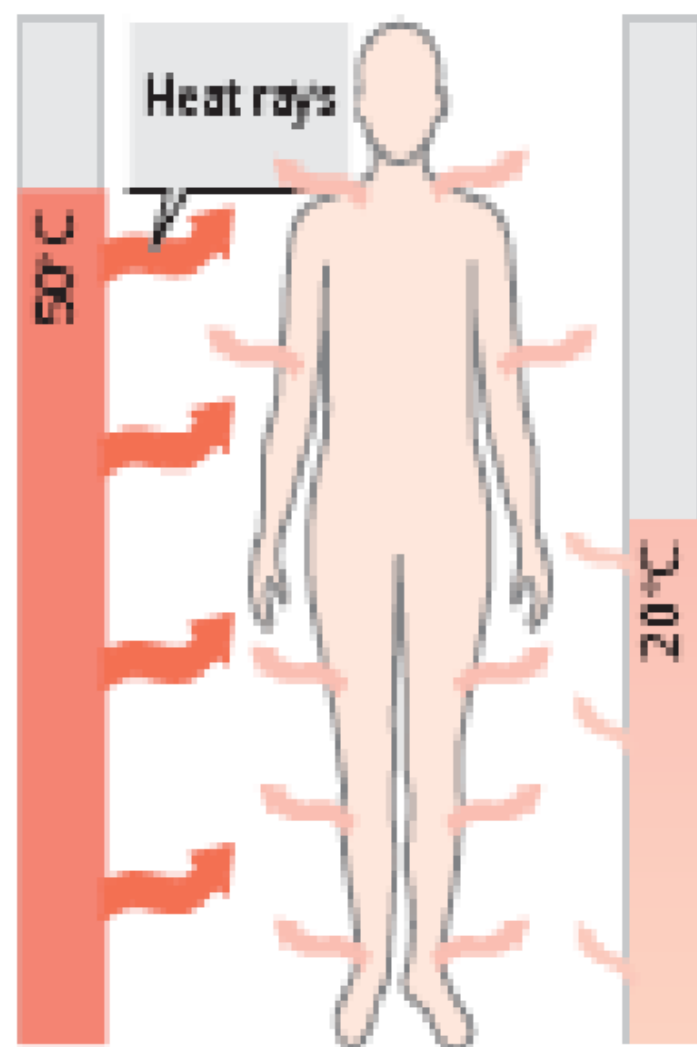
- **Ingestion of food.** The ingestion of food raises the metabolic rate due to the energy “costs” of digesting, absorbing, and storing nutrients.
- **Age.** The metabolic rate of a child, in relation to its size, is about double that of an elderly person due to the high rates of reactions related to growth.
- **Other factors.** Other factors that affect metabolic rate are gender (lower in females, except during pregnancy and lactation), climate (lower in tropical regions), sleeping (lower), and malnutrition (lower).

Heat Loss (Thermolysis)

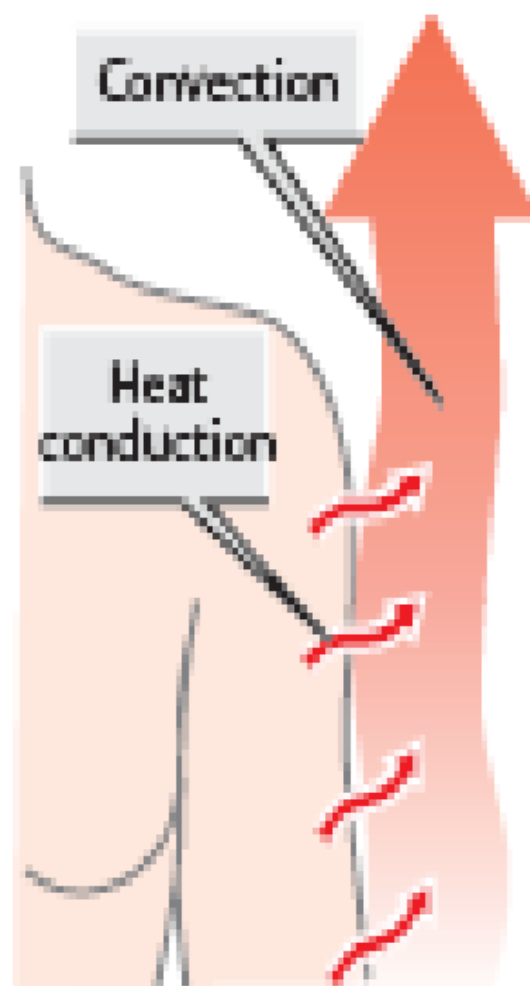
- Radiation
- Conduction
- Convection
- Evaporation
 - Perspiration
 - Respiration
- Loss through urine & feces

B. Mechanisms of heat loss

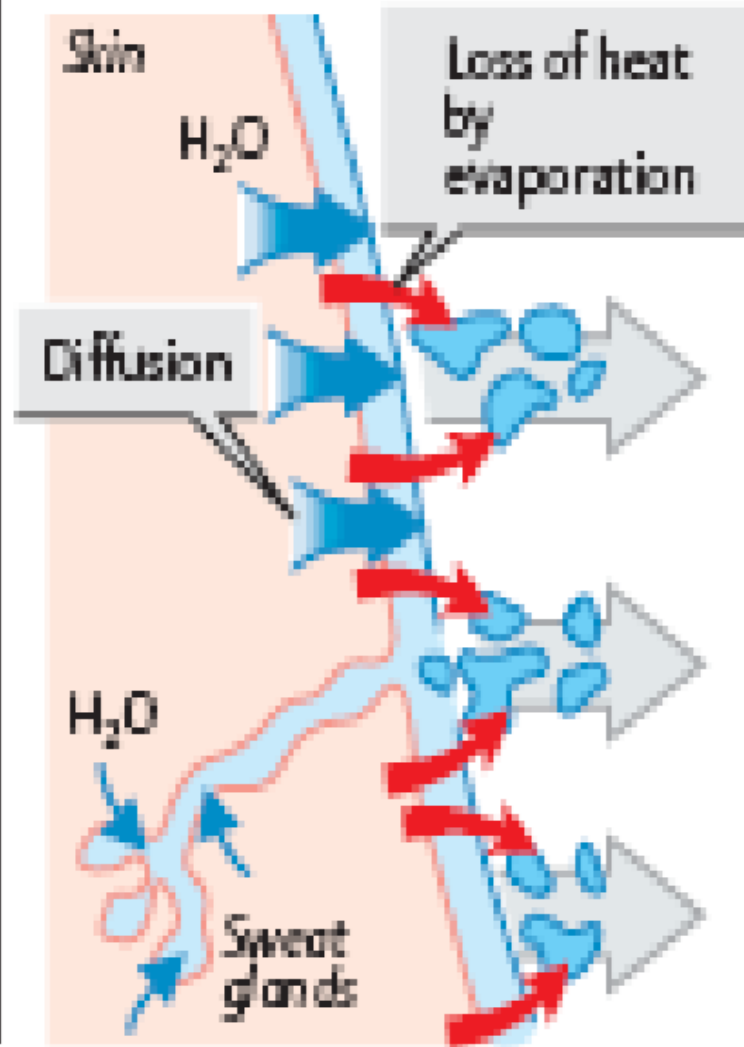
1 Radiation



2 Conduction and convection



3 Evaporation



Heat Loss (Thermolysis) continued..

1. Conduction is the heat exchange that occurs between molecules of two materials that are in direct contact with other. At rest, about 3% of body heat is lost via conduction to solid materials in contact with the body, such as a chair, clothing, and jewelry. Heat can also be gained via conduction—for example, while soaking in a hot tub.

2. Convection is the transfer of heat by the movement of a fluid (a gas or a liquid) between areas of different temperature. The contact of air or water with body results in heat transfer by both conduction and convection. At rest, about 15% of body heat is lost to the air via conduction and convection.

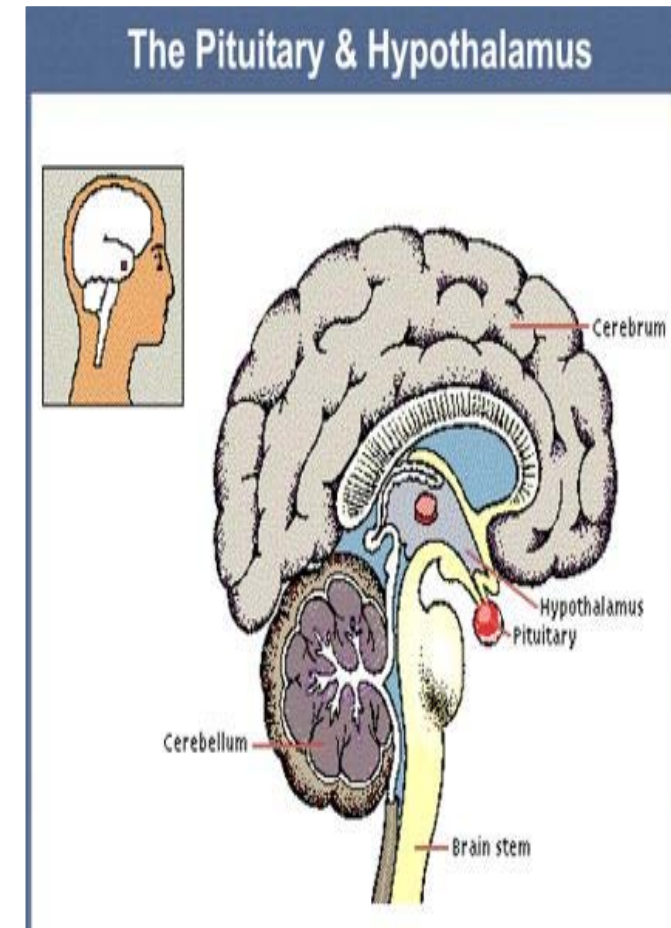
Heat Loss (Thermolysis) continued..

3. Radiation is the transfer of heat in the form of infrared rays between a warmer object and a cooler one without physical contact, In a room at 21C, about 60% of heat loss occurs via radiation in a resting person.

4. Evaporation is the conversion of a liquid to a vapor, Under typical resting conditions, about 22% of heat loss occurs through evaporation of about 700 mL of water per day—300 mL in exhaled air and 400 mL from the skin surface. it is termed **insensible water loss**.

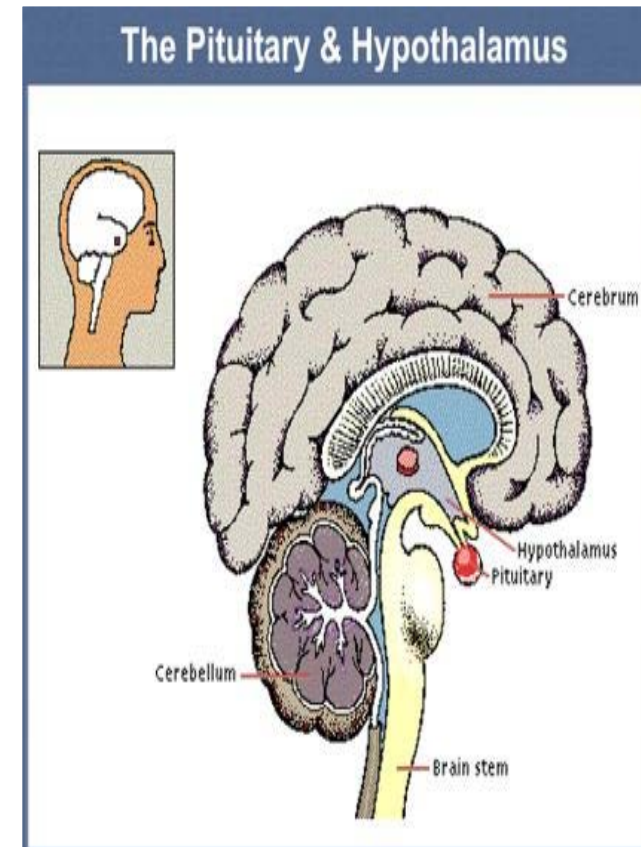
Thermoregulation via Hypothalamic Thermostat

- Temperature is regulated by nervous feedback mechanisms.
- Thermoregulatory center located in the **Hypothalamus**, it is called **preoptic area**.
- The preoptic area receives impulses from thermoreceptors in the skin and mucous membranes (**cutaneous thermoreceptors**) and in the hypothalamus (**central thermoreceptors**).
- central thermoreceptors detect changes in blood temperature.



Thermoregulation via Hypothalamic Thermostat

- Temperature is regulated via a Negative Feedback Mechanism.
- Nerve impulses from the **preoptic area** propagate to two other parts of the hypothalamus known as the **heat-losing center** and the **heat-promoting center**, which, when stimulated by the preoptic area, set into operation a series of responses that lower body temperature and raise body temperature, respectively.



Thermoregulation via Negative Feedback Mechanism:

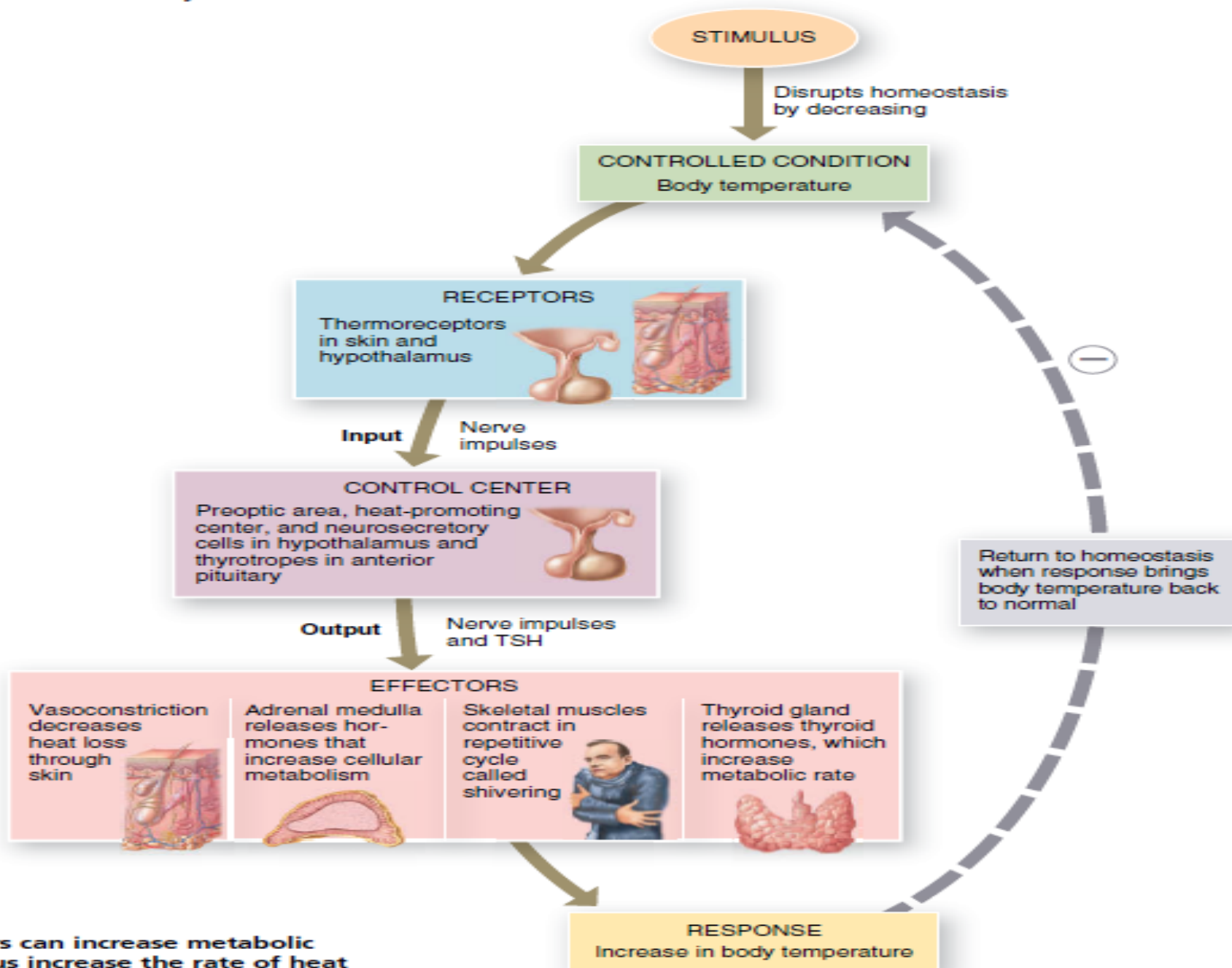
If core temperature declines, mechanisms that help conserve heat and increase heat production act via several negative feedback loops:

1. Thyroid Hormones Releasing:

- Thermoreceptors in the skin and hypothalamus send nerve impulses to the preoptic area and the heat-promoting center in the hypothalamus, and to produce thyrotropin-releasing hormone (TRH).
- TRH stimulates anterior pituitary to release thyroid-stimulating hormone (TSH).
- The thyroid gland responds to TSH by releasing more thyroid hormones into the blood. As increased levels of thyroid hormones slowly increase the metabolic rate, body temperature rises.



Core temperature is the temperature in body structures deep to the skin and subcutaneous layer; shell temperature is the temperature near the body surface.

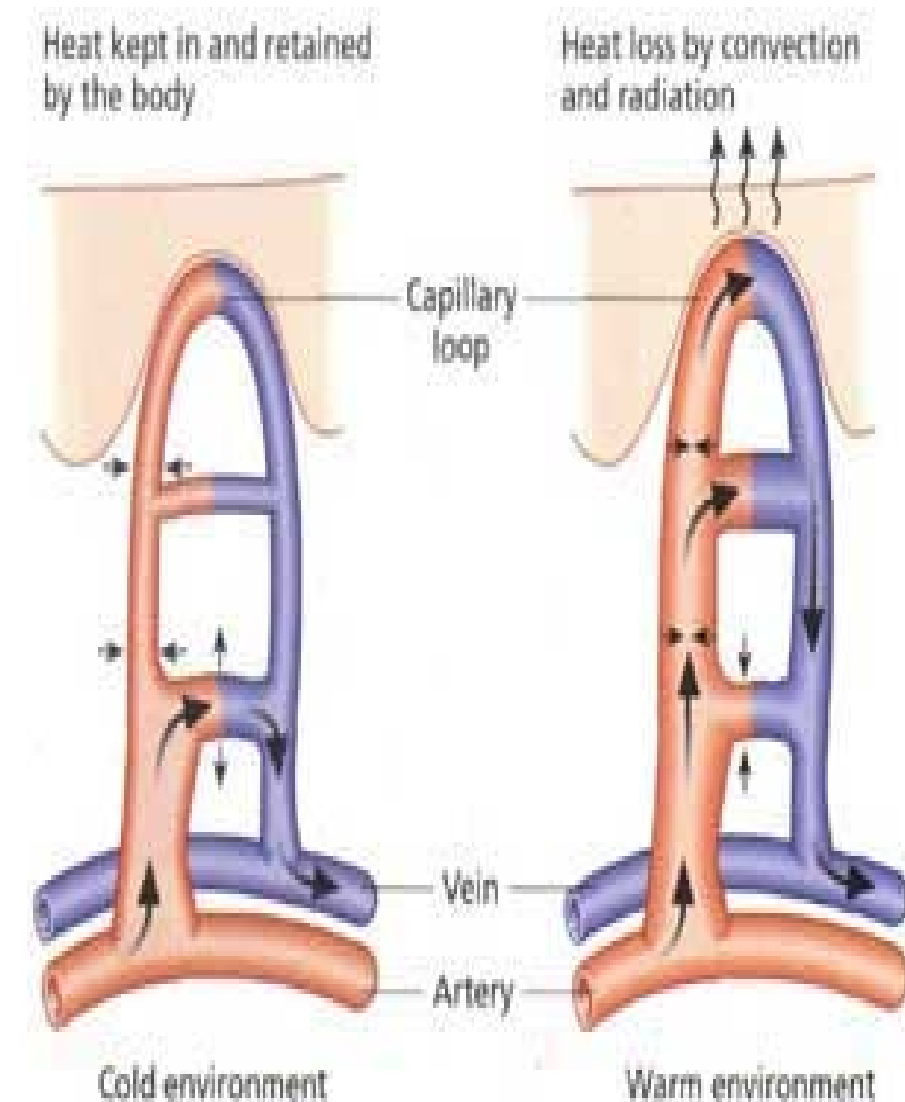


What factors can increase metabolic rate and thus increase the rate of heat production?

Thermoregulation via Negative Feedback Mechanism

2. Vasoconstriction:

- Nerve impulses from the heat-promoting center stimulate sympathetic nerves that cause blood vessels of the skin to constrict.
- Vasoconstriction decreases the flow of warm blood, and thus decreases the transfer of heat, from the internal organs to the skin.



Thermoregulation via Negative Feedback Mechanism

3. Epinephrine and Norepinephrine Releasing:

- Nerve impulses in sympathetic nerves leading to the adrenal medullae stimulate the release of epinephrine and norepinephrine into the blood.
- The hormones in turn bring about an increase in cellular metabolism, which increases heat production.

Thermoregulation via Negative Feedback Mechanism

4. Muscle Contraction (Shivering):

- The heat-promoting center stimulates parts of the brain that increase muscle tone in a repetitive cycle called (Shivering).
- Shivering greatly increases the rate of heat production. During maximal shivering, body heat production can rise to about four times the basal rate in just a few minutes.

5. Others - **Behavioral Control of Temperature** (psychic sensation of being cold) e.g making appropriate environmental or behavioral adjustments to re-establish comfort such as moving to a heated area or Curling up.

- **Piloerection or Horripilation** (Arrector Pili muscle contraction) to trap warm air, it give the feeling of 'goose bumps'



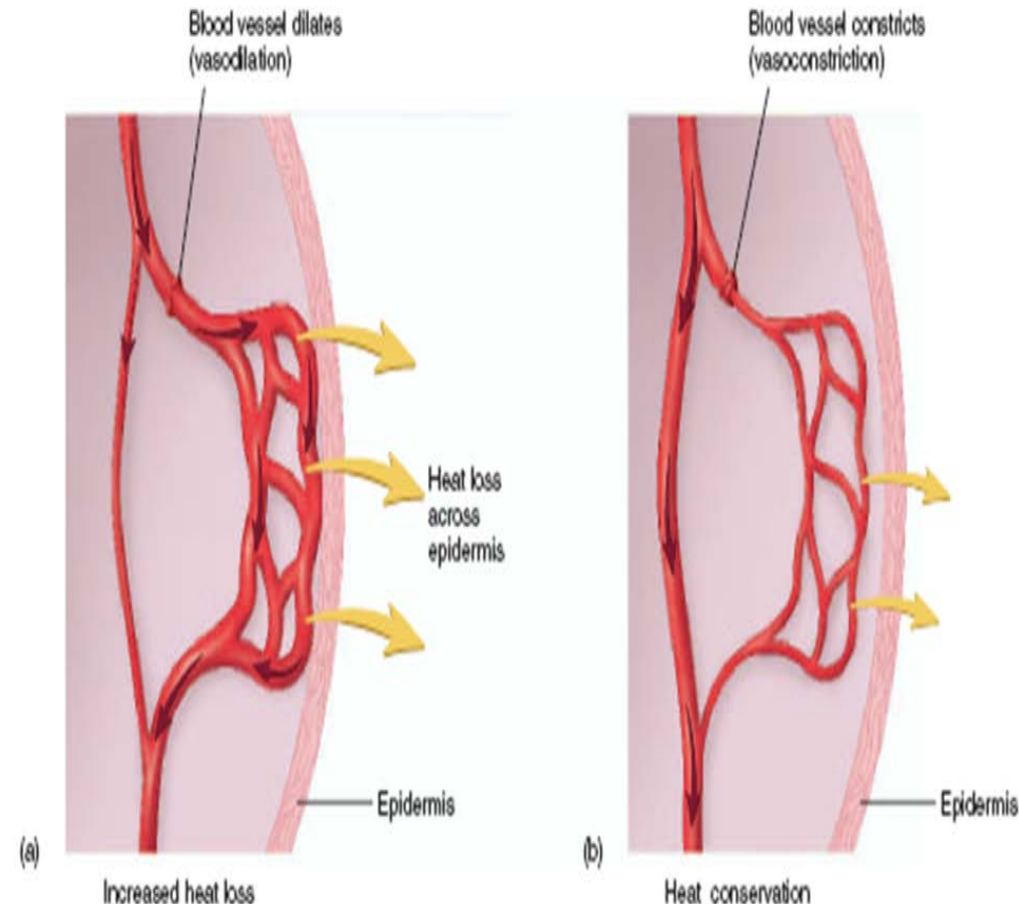
Thermoregulation via Negative Feedback Mechanism

If core body temperature rises above normal, negative feedback lead to:

1. Vasodilatation:

Nerve impulses from the heat-losing center cause dilation of blood vessels in the skin.

The skin becomes warm, and the excess heat is lost to the environment via radiation and conduction.

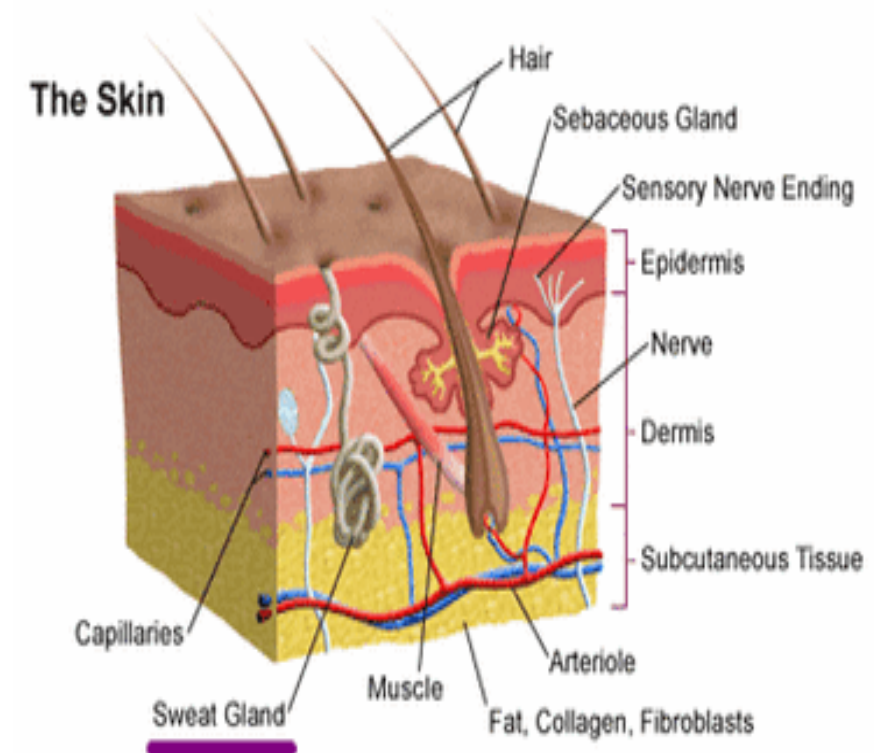


Thermoregulation via Negative Feedback Mechanism

2. Metabolic rate decreases, and shivering does not occur.

3. Sweating and Evaporation

As the water in perspiration evaporates from the surface of the skin, the skin is cooled.



*Questions and
comments?*

